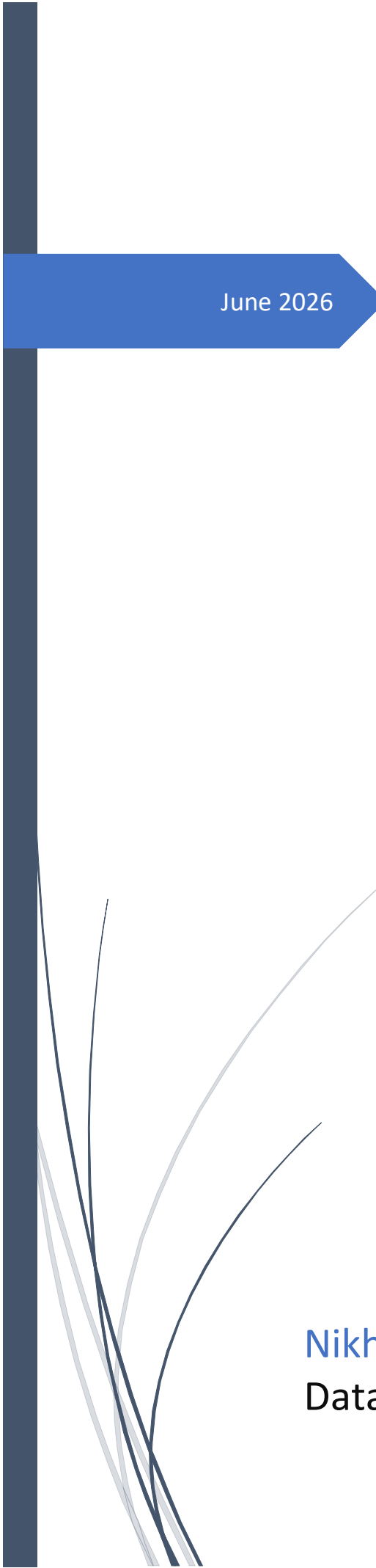




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Healthcare Analytics

End-to-End Data Analysis Using Python, SQL & Power BI

Dataset Source: Kaggle – Prasad Patil
Tools Used: Python | SQL | Power BI

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Data Analytics Portfolio Project

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1. Project Overview

This Healthcare Analytics project presents an end-to-end data analysis of a synthetic healthcare dataset containing information about patient demographics, medical conditions, hospital admissions, billing amounts, insurance providers, medications, and test results.

The project follows a complete data analytics workflow using Python, SQL, and Power BI. Python was used to inspect, clean, transform, and explore the dataset, including additional variables to support further analysis. SQL was then used to answer structured analytical questions related to patient admissions, medical conditions, healthcare billing, length of stay, insurance providers, patient age groups, and admission trends. Finally, Power BI was used to develop an interactive dashboard that presents the main findings through key performance indicators (KPIs), visualisations, and interactive slicers.

The purpose of the project is to identify and communicate patterns in patient activity, healthcare costs, hospital utilisation, medical conditions, and test outcomes while demonstrating the ability to transform raw healthcare data into meaningful analytical insights and an interactive business intelligence solution.

2. Business Problem and Objective

2.1. Business Problem

Healthcare organisations generate large volumes of data related to patients, hospital admissions, medical conditions, treatment outcomes, billing, and insurance. However, raw healthcare data alone provides limited value unless it is properly cleaned, analysed, and transformed into meaningful information.

Understanding patterns in-patient admissions, length of stay, medical conditions, and healthcare costs can support better operational and financial analysis. Without a structured analytical approach, it can be difficult to identify which conditions account for higher patient volumes, whether hospital utilisation differs across admission types, how healthcare costs vary across patient groups, and how admission activity changes over time. This project therefore uses healthcare data to explore these areas and demonstrate how analytical techniques can transform patient-level records into clear and actionable insights.

2.2. Project Objective

The objective of this project is to analyse the healthcare dataset to identify patterns and relationships across patient demographics, medical conditions, hospital admissions, length of stay, healthcare billing, insurance providers, test results, and admission trends over time.

Specifically, the project aims to:

- a. Examine the distribution of patients across medical conditions, admission types, demographics, and test results.
- b. Analyse hospital length of stay and determine how it varies across admission types.
- c. Compare healthcare billing across medical conditions, insurance providers, and admission categories.
- d. Examine patient admission patterns and changes over time.
- e. Use Python for data cleaning, adding variables, and exploratory data analysis.

- f. Use SQL to answer structured analytical questions and perform aggregation, segmentation, comparison, and ranking.
- g. Develop an interactive Power BI dashboard that allows users to monitor key healthcare metrics and explore the data using dynamic filters.

3. Dataset Overview

The dataset used in this project is the Healthcare Dataset published by Prasad Patil on Kaggle. It is a synthetic healthcare dataset designed for data analysis and data science practice. The dataset contains patient-level information covering demographics, medical conditions, hospital admissions, healthcare providers, insurance, billing, medication, and test results.

Dataset Source: [Healthcare Dataset — Prasad Patil, Kaggle](#)

The original dataset contained 55,500 records and 15 variables. During the data-cleaning stage, duplicate records and billing-related data quality issues were identified and handled. Following the cleaning process, the final dataset contained 54,966 records.

3.1 Dataset Variable

The original dataset contained following 15 Variables:

Variable	Description
Name	Patient name
Age	Patient age
Gender	Patient gender
Blood Type	Patient blood group
Medical Condition	Diagnosed medical condition
Date of Admission	Date the patient was admitted
Doctor	Doctor responsible for the patient
Hospital	Hospital where the patient was treated
Insurance Provider	Patient's insurance provider
Billing Amount	Amount billed for healthcare services
Room Number	Hospital room assigned to the patient
Admission Type	Type of admission: Emergency, Elective, or urgent
Discharge Date	Date the patient was discharged
Medication	Medication prescribed to the patient
Test Results	Result of the patient's medical test

3.2 Additional Variables

During data preparation in Python, additional features were created to support the analysis:

- a. Length of Stay — calculated using the difference between Discharge Date and Date of Admission.
- b. Billing Valid — created to identify valid billing records during the billing-quality checks.
- c. Admission Year — extracted from the admission date for yearly analysis.
- d. Admission Month — extracted for monthly analysis.
- e. Admission Year-Month — created to support chronological analysis of patient admissions.

These additional variables were later used across the Python analysis, SQL queries, and Power BI dashboard.

3.3 Dataset Consideration

An important consideration is that the dataset is synthetic rather than real patient data. Therefore, the findings from this project should be interpreted as a demonstration of healthcare data analytics techniques rather than conclusions about actual hospitals or patient populations. This is particularly relevant because several variables in the dataset are distributed quite evenly across categories, which should be considered when interpreting the analytical results.

4. Tools & Technologies

The project used Python, SQL, and Microsoft Power BI to complete different stages of the data analytics workflow, from data preparation and exploratory analysis to structured querying and interactive visualisation.

4.1 Python – Google Colab

Python was used as the primary tool for data cleaning, transformation, and exploratory data analysis (EDA). This analysis was performed in Google Colab, providing a notebook-based environment for working with the healthcare dataset.

The main Python libraries used were:

- a. Pandas – for loading, inspecting, cleaning, transforming, and analysing the dataset.
- b. Matplotlib – for creating visualisations during exploratory data analysis.

Python was also used to convert data fields, calculate Length of Stay, create time-based features, investigate billing-related data issues, remove duplicate records, and export the cleaned dataset for subsequent analysis.

4.2 SQL- SQLite/ DB Browser for SQLite

After data preparation was completed, the cleaned dataset was imported into SQLite using DB Browser for SQLite.

SQL was used to answer structured analytical questions involving:

- a. Patient counts and medical conditions
- b. Admission types
- c. Length of stay
- d. Healthcare billing
- e. Insurance providers
- f. Patient age groups
- g. Admission trends over time

The SQL analysis applied techniques including filtering, aggregation, GROUP BY, HAVING, conditional logic, Common Table Expressions (CTEs), and window functions. For example, the project used a CTE

to compare admission-type billing with the overall average and RANK() to rank medical conditions by average billing.

4.3 Microsoft Power BI

Microsoft Power BI was used to transform the cleaned and analysed data into an interactive healthcare analytics dashboard.

DAX measures were created for four key performance indicators:

- a. Total Patients
- b. Average Billing
- c. Average Length of Stay
- d. Total Billing

The dashboard also included visualisations covering medical conditions, admission types, test results, length of stay, insurance providers, healthcare billing, and admission trends over time. Interactive slicers for Admission Year, Medical Condition, and Insurance Provider allow users to dynamically filter the dashboard and explore different segments of the data.

5. Data Cleaning & Preparation

Before conducting the analysis, the healthcare dataset was inspected and cleaned in Python using Pandas to identify data quality issues and prepare the data for further analysis in Python, SQL, and Power BI. The initial dataset contained 55,500 records and 15 columns. The cleaning process focused on duplicate records, data types, date fields, billing values, and the creation of additional variables required for the analysis.

5.1 Initial Data Inspection

The dataset was first examined to understand its structure, data types, numerical distributions, categorical values, and overall data quality. Functions such as head(), shape, info(), describe(), and missing-value checks were used during the initial inspection. The initial review showed that the dataset contained 55,500 records and 15 variables, with no missing values in the original data.

5.2 Duplicate Records

Duplicate records were investigated to ensure that repeated observations did not affect patient counts or analytical results. A total of 534 exact duplicate records were identified and removed. After duplicate removal, the dataset contained 54,966 records, with no exact duplicates remaining.

5.3 Date Validation and Conversion

The Date of Admission and Discharge Date columns were converted from object/text format to the appropriate datetime format. The dates were also validated to ensure that discharge dates occurred after admission dates. No negative or zero-length hospital stays were identified. A new variable, Length of Stay, was calculated as the difference between Discharge Date and Date of Admission. Length of stay ranged from 1 to 30 days, with an overall average of approximately 15.5 days.

5.4 Billing Data Validation

The Billing Amount column was examined for invalid values. The analysis identified 106 records with negative billing amounts, which were considered invalid for billing analysis. Rather than converting these negative amounts into positive values, a Billing Valid indicator was created to identify valid and invalid billing records. The negative billing amounts were then replaced with missing (NaN) values so that they would not distort billing calculations. This approach preserved the patient records while excluding invalid billing values from calculations that required valid healthcare costs.

5.5 Data Transformation

Additional variables were created from the existing data to support subsequent analysis:

- a. Length of Stay – number of days between admission and discharge.
- b. Billing Valid – indicator used to distinguish valid and invalid billing records.
- c. Admission Year – year extracted from Date of Admission.
- d. Admission Month – month extracted from Date of Admission.
- e. Admission Year-Month – combined year and month used for chronological admission analysis.

These variables supported the later EDA, SQL analysis, and Power BI dashboard.

5.6 Final Data Quality Check

After completing the cleaning and preparation process, a final data-quality check was performed.

The final dataset contained:

- a. 54,966 records
- b. No exact duplicate records
- c. No negative billing amounts remaining
- d. No negative or zero Length of Stay values
- e. Correct datetime formats for admission and discharge dates
- f. Additional analytical variables required for subsequent analysis

The cleaned dataset was then exported as healthcare_cleaned.csv for use in the SQL and Power BI stages of the project.

6. Exploratory Data Analysis (EDA)

After completing the data cleaning and preparation process, Exploratory Data Analysis (EDA) was conducted in Python to better understand the characteristics and patterns within the healthcare dataset. The analysis focused on patient demographics, medical conditions, admission patterns, length of stay, and healthcare billing.

6.1 Patient Demographics

Patient demographics were analysed using age, gender, and blood type. The gender distribution was almost evenly balanced, with 27,496 male patients (50.02%) and 27,470 female patients (49.98%). This indicates that neither gender is significantly overrepresented in the dataset. Patient ages ranged

from 13 to 89 years, with an average age of approximately 51.5 years and a median age of 52 years, indicating a broad age distribution across the dataset. Blood types were also distributed very evenly. Each of the eight blood groups represented approximately 12.4%–12.6% of patient records, with no individual blood type dominating the dataset.

6.2 Medical Conditions

The distribution of patients across the six medical conditions was examined to identify the most frequently represented conditions. Arthritis had the highest number of patient records with 9,218, closely followed by Diabetes (9,216) and Hypertension (9,151). Asthma had the lowest count with 9,095 records. However, the differences between all six conditions were relatively small, indicating that medical conditions are distributed quite evenly throughout the dataset.

6.3 Admission Patterns and Length of Stay

Patient admissions were analysed across Elective, Urgent, and Emergency admission types.

Elective admissions were the most common with 18,473 patients (33.61%), followed by Urgent admissions with 18,391 (33.46%) and Emergency admissions with 18,102 (32.93%). The small differences indicate a nearly even distribution across the three admission categories.

Average Length of Stay was also compared across admission types:

- a. Emergency: 15.58 days
- b. Elective: 15.51 days
- c. Urgent: 15.40 days

Although Emergency admissions had the longest average stay, the differences were minimal. This suggests that admission type is not associated with substantial differences in Length of Stay within this dataset.

6.4 Healthcare Billing

Average billing amounts were compared across medical conditions to identify potential differences in healthcare costs. Obesity recorded the highest average billing amount at approximately \$25,859.22, followed by Diabetes at \$25,714.33. Cancer recorded the lowest average billing amount at approximately \$25,205.92. Despite these differences, the overall range between medical conditions was relatively narrow. This suggests that medical condition alone does not appear to be a strong source of billing variation within this synthetic dataset.

6.5 EDA Summary

The exploratory analysis showed that several major categorical variables—including gender, blood type, medical condition, and admission type—were distributed relatively evenly. Patients represented a broad range of ages, while the average hospital stay was approximately 15.5 days. Billing amounts showed some differences across medical conditions, although the variation was relatively limited. Overall, the EDA provided an initial understanding of the cleaned dataset and established a foundation for the more structured analytical questions investigated using SQL.

7. SQL Analysis

After completing data cleaning and exploratory analysis in Python, the cleaned healthcare dataset was imported into SQLite using DB Browser for SQLite. SQL was then used to answer structured analytical questions and further investigate patterns identified during the exploratory analysis. The analysis included patient segmentation, aggregation, billing comparisons, ranking, and time-based analysis.

7.1 Medical Conditions and Patient Distribution

SQL was used to determine which medical conditions were most frequently represented in the dataset. Arthritis had the highest patient count with 9,218 records, followed closely by Diabetes (9,216) and Hypertension (9,151). Asthma had the lowest count with 9,095 records. The small differences between the six conditions confirm that medical conditions are relatively evenly distributed throughout the dataset.

7.2 Admission Types and Length of Stay

Patient volumes were compared across the three admission types. Elective admissions were the most frequent with 18,473 records, followed by Urgent (18,391) and Emergency (18,102). Average Length of Stay was then calculated for each admission type. Emergency admissions recorded the highest average stay at 15.58 days, followed by Elective at 15.51 days and Urgent at 15.40 days. The differences were minimal, suggesting that admission type is not associated with substantial differences in Length of Stay within this dataset.

7.3 Billing by Medical Condition

Average healthcare billing was compared across medical conditions. Obesity recorded the highest average billing amount at \$25,859.22, followed by Diabetes at \$25,714.33. Cancer recorded the lowest average billing amount at \$25,205.92. Although differences were present, the overall range was relatively narrow, indicating that average billing was fairly consistent across medical conditions. A HAVING clause was also used to identify conditions with an average billing amount above \$25,500, demonstrating how SQL can filter aggregated results rather than individual records.

7.4 Insurance Provider Analysis

Average billing was analysed across insurance providers. Medicare recorded the highest average billing amount at approximately \$25,678.09, while UnitedHealthcare recorded the lowest at approximately \$25,458.89. However, the difference between the highest and lowest averages was relatively small, indicating broadly similar average billing levels across insurance providers.

7.5 Patient Age Group Analysis

SQL CASE logic was used to segment patients into five age groups:

Under 18, 18–35, 36–50, 51–65, and 66+.

The 66+ group was the largest with 16,096 patients, followed by the 18–35 group with 14,289. The 51–65 and 36–50 groups contained 12,298 and 12,167 patients respectively, while only 116 patients

were under 18. This analysis demonstrates the use of conditional logic to transform continuous age values into meaningful analytical categories.

7.6 Admission Billing Compared with the Overall Average

A Common Table Expression (CTE) was used to calculate the overall average billing amount and compare it with average billing across admission types. Elective admissions recorded an average billing amount of \$25,663.34, slightly above the overall dataset average of \$25,594.63. Urgent and Emergency admissions did not exceed the overall average. The small difference indicates that billing amounts were relatively similar across admission categories.

7.7 Ranking Medical Conditions by Average Billing

A SQL window function using RANK() was applied to rank medical conditions according to their average billing amount. Obesity ranked first, followed by Diabetes and Asthma, while Cancer ranked last with the lowest average billing amount. This analysis extended the basic aggregation results by producing an ordered ranking of medical conditions based on healthcare costs.

7.8 Admission Trend Over Time

Monthly patient admissions were analysed using the Admission Year-Month variable. Admissions remained relatively stable across most complete months, generally ranging between approximately 850 and 1,000 admissions per month, with no clear sustained upward or downward trend. The lower admission counts in May 2019 and May 2024 should be interpreted carefully because these represent partial periods within the dataset rather than necessarily indicating a genuine reduction in patient activity.

7.9 SQL Analysis Summary

Overall, the SQL analysis reinforced several patterns identified during Python EDA while providing a more structured approach to answering analytical questions. The analysis demonstrated the use of SQL for aggregation, filtering, segmentation, comparison, conditional logic, ranking, and time-based analysis. These results were subsequently incorporated into the development of the interactive Power BI dashboard.

8. Power BI Dashboard

After completing the Python and SQL analysis, the cleaned healthcare dataset was imported into **Microsoft Power BI** to develop an interactive healthcare analytics dashboard. The dashboard was designed to bring the main analytical findings together in a single view, allowing users to monitor key healthcare metrics and explore patterns across medical conditions, admission types, billing, insurance providers, test results, and patient admissions over time.

8.1 Dashboard Overview

The dashboard provides a high-level overview of the healthcare dataset while also allowing users to explore individual areas of interest through interactive filtering. It combines four key performance indicators (KPIs), seven analytical visualisations, and three interactive slicers. Together, these

elements provide information about patient volume, healthcare costs, hospital utilisation, medical conditions, admission characteristics, insurance providers, test outcomes, and admission trends.

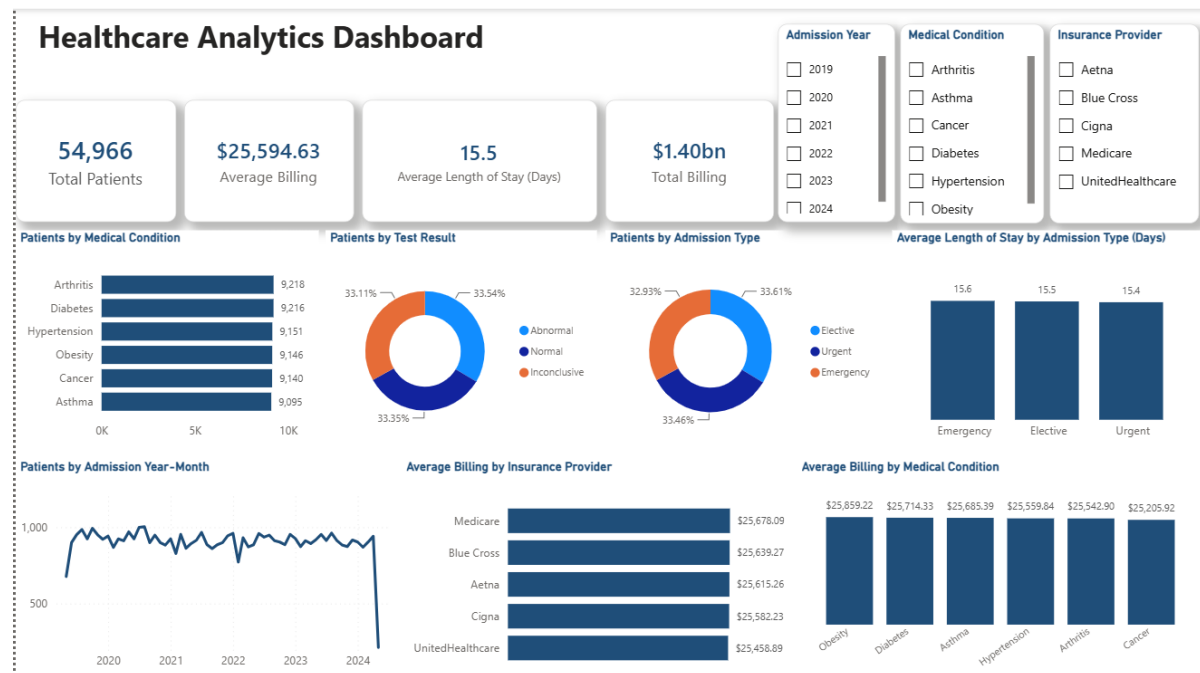


Figure 1. Healthcare Analytics – Interactive Power BI Dashboard

8.2 Key Performance Indicators

Four DAX measures were created to provide an immediate summary of the dataset:

- Total Patients – 54,966: represents the total number of patient records in the cleaned dataset.
- Average Billing – \$25,594.63: represents the average valid healthcare billing amount.
- Average Length of Stay – 15.5 days: represents the average number of days patients remained in hospital.
- Total Billing – approximately \$1.40 billion: represents the total value of valid healthcare billing across the dataset.

These KPIs allow users to quickly understand the overall scale of patient activity, healthcare costs, and hospital utilisation before exploring more detailed visualisations.

8.3 Dashboard Visualisations

Seven visualisations were included to examine different aspects of the healthcare data:

- Patients by Medical Condition – compares patient volumes across the six medical conditions.
- Patients by Test Result – shows the distribution of Abnormal, Normal, and Inconclusive test results.
- Patients by Admission Type – compares the proportion of Emergency, Elective, and Urgent admissions.
- Average Length of Stay by Admission Type (Days) – compares hospital utilisation across the three admission categories.
- Patients by Admission Year-Month – displays changes in patient admission volumes over time.

- f. Average Billing by Insurance Provider – compares average healthcare billing across insurance providers.
- g. Average Billing by Medical Condition – compares average healthcare costs across the six medical conditions.

The visualisations allow both operational and financial patterns to be examined within the same dashboard. The completed analysis showed, for example, relatively balanced admission types and test-result distributions, while billing differences across medical conditions and insurance providers were comparatively small.

8.4 Interactive Slicers

Three interactive slicers were incorporated into the dashboard:

- a. Admission Year
- b. Medical Condition
- c. Insurance Provider

These slicers allow users to filter the dashboard and investigate specific subsets of the healthcare data. When a filter is selected, the KPIs and relevant visualisations dynamically update to reflect the selected population. During the final dashboard testing, the slicers were checked individually and in combination to confirm that the KPIs and visualisations responded correctly to user selections.

8.5 Power BI Summary

The Power BI dashboard represents the final visualisation stage of the project's end-to-end analytics workflow. It transforms the cleaned and analysed healthcare data into an interactive format that enables users to move from a high-level overview to more detailed exploration. By combining DAX measures, KPI cards, analytical charts, time-series analysis, and interactive slicers, the dashboard provides a clear and accessible way to communicate the main findings generated throughout the project.

9. Key Findings & Insights

The combined analysis conducted using Python, SQL, and Power BI identified several key patterns within the healthcare dataset.

9.1 Balanced Patient Demographics

The dataset showed an almost equal gender distribution, with 50.02% male and 49.98% female patient records. Patient ages ranged from 13 to 89 years, with an average age of approximately 51.5 years. SQL-based age segmentation further showed that the 66+ age group was the largest, containing 16,096 patient records, while only 116 records represented patients under 18.

9.2 Medical Conditions Were Evenly Distributed

Patient volumes across the six medical conditions were relatively similar. Arthritis was the most common condition with 9,218 records, while Asthma was the least common with 9,095. The difference

of only 123 records between the highest and lowest categories indicates that no individual medical condition strongly dominates the dataset.

9.3 Admission Patterns and Length of Stay Were Consistent

Admission volumes were also relatively balanced. Elective admissions were the most common with 18,473 records, followed closely by Urgent and Emergency admissions, with each admission type representing approximately one-third of the dataset. Length of Stay showed very little variation across admission types. Emergency admissions had the highest average at 15.58 days, compared with 15.51 days for Elective and 15.40 days for Urgent admissions. This suggests that, within this dataset, admission type alone does not appear to be strongly associated with differences in hospital stay duration.

9.4 Healthcare Billing Showed Limited Variation

Average billing varied only moderately across medical conditions. Obesity recorded the highest average billing amount at approximately \$25,859, while Cancer recorded the lowest at approximately \$25,206. A similar pattern was observed across insurance providers. Medicare recorded the highest average billing at approximately \$25,678, while UnitedHealthcare recorded the lowest at approximately \$25,459. The relatively narrow differences suggest that neither medical condition nor insurance provider produced substantial variation in average billing within this dataset.

9.5 Test Results Were Almost Evenly Distributed

The three test-result categories—Abnormal, Normal, and Inconclusive—each represented approximately one-third of patient records. This relatively balanced distribution means that no single test-result category dominates the dataset.

9.6 Patient Admission Remained Relatively Stable Over Time

Monthly patient admissions were generally stable across the complete periods covered by the dataset, typically ranging between approximately 800 and 1,000 admissions per month. Lower admission counts at the beginning and end of the dataset should be interpreted carefully because these periods contain partial data coverage, rather than necessarily representing a genuine decline in patient demand.

9.7 Overall Insight

A recurring pattern throughout the analysis is the relatively even distribution of many variables, including gender, medical conditions, admission types, test results, and several billing comparisons. Because the healthcare dataset is synthetic, these balanced patterns are likely influenced by how the dataset was generated. Therefore, the results should primarily be viewed as a demonstration of data cleaning, exploratory analysis, SQL analysis, and business intelligence techniques, rather than evidence about real-world healthcare populations or hospital performance.

10. Business Recommendations

Based on the patterns identified throughout the analysis, several areas could be considered for further investigation and decision-making if similar trends were observed in real healthcare data.

10.1 Monitor Length of Stay Across Admission Types

Emergency admissions recorded the highest average Length of Stay at 15.58 days, compared with 15.51 days for Elective and 15.40 days for Urgent admissions. However, the differences between the three categories were relatively small. Healthcare organisations should therefore avoid assuming that admission type alone explains hospital utilisation. Further analysis could incorporate factors such as patient age, medical condition, hospital, treatment, and discharge patterns to better understand what contributes to longer hospital stays.

10.2 Investigate the Drivers of Healthcare Billing

Average billing showed only limited variation across both medical conditions and insurance providers. Obesity recorded the highest average billing among medical conditions at approximately \$25,859, while Medicare recorded the highest average among insurance providers at approximately \$25,678. Rather than relying on medical condition or insurance provider alone, future cost analysis should investigate combinations of variables such as Length of Stay, admission type, patient age, medication, hospital, and medical condition to better understand the factors associated with healthcare costs.

10.3 Use Age Segmentation for Further Patient Analysis

The SQL analysis identified the 66+ age group as the largest segment, containing 16,096 patient records. In a real healthcare setting, age segmentation could be incorporated into operational reporting to examine whether different age groups experience differences in medical conditions, Length of Stay, admission types, billing, and test outcomes. This could provide a more detailed understanding of patient utilisation across age groups.

10.4 Monitor Admission Trends Over Time

Monthly admissions were relatively stable during complete periods of the dataset, generally ranging between approximately 800 and 1,000 patients per month. The lower values at the beginning and end of the dataset resulted from partial data coverage and should therefore not automatically be interpreted as declining patient demand. Healthcare organisations using real operational data should continuously monitor admission trends over time and distinguish between genuine changes in patient activity and incomplete reporting periods before making staffing, capacity, or resource-planning decisions.

10.5 Use Interactive Dashboards for Ongoing Monitoring

The Power BI dashboard enables users to explore the data using Admission Year, Medical Condition, and Insurance Provider slicers, with the KPIs and relevant visualisations dynamically updating according to the selected filters. In a real-world environment, a similar dashboard could support ongoing monitoring of patient volumes, healthcare costs, hospital utilisation, and admission trends. Expanding the dashboard with regularly refreshed operational data could help decision-makers identify changes and investigate areas requiring further attention.

10.6 Validate Findings Using Real-World Healthcare Data

Because the dataset used in this project is synthetic and several variables are unusually evenly distributed, the findings should not be used directly to guide real healthcare decisions. Before applying

any of the identified patterns operationally, the analysis should be replicated using real, appropriately governed and de-identified healthcare data. This would allow the relationships observed in this project to be tested under realistic patient and hospital conditions.

11. Project Limitations

Although the project demonstrates a complete data analytics workflow using Python, SQL, and Power BI, several limitations should be considered when interpreting the results.

11.1 Synthetic Dataset

The primary limitation is that the healthcare dataset is synthetic rather than real-world patient data. As a result, the patterns identified in the analysis may not accurately represent actual patient populations, healthcare costs, hospital operations, or clinical outcomes. Several variables—including medical conditions, admission types, gender, blood types, and test results—were distributed relatively evenly throughout the dataset. This suggests that some of the observed patterns may reflect the way the synthetic data was generated rather than naturally occurring healthcare behaviour.

11.2 Limited Variables

The dataset contains useful information about patient demographics, admissions, medical conditions, billing, insurance, medications, and test results, but it does not capture all factors that could influence healthcare outcomes or costs. Additional variables such as treatment procedures, diagnosis severity, comorbidities, staffing levels, hospital capacity, and detailed clinical outcomes could provide a more comprehensive understanding of hospital utilisation and healthcare costs.

11.3 Data Quality and Billing Values

During data preparation, 106 negative billing values were identified as invalid and excluded from billing calculations while preserving the associated patient records. Although this approach prevented invalid values from affecting financial analysis, it also means that billing-based calculations were performed using fewer valid billing observations than the total number of patient records.

11.4 Partial Time Periods

The dataset covers admissions from May 2019 through May 2024. The beginning and ending periods do not represent complete calendar years, which affects comparisons involving those periods. For this reason, lower admission counts at the beginning and end of the time series should not automatically be interpreted as genuine decreases in patient activity.

11.5 Analytical Scope

The project primarily uses descriptive and exploratory analysis to identify patterns, comparisons, and trends within the dataset. Therefore, relationships observed between variables should not be interpreted as evidence of causation. More advanced statistical analysis or predictive modelling would be required to investigate whether particular factors significantly influence outcomes such as Length of Stay or healthcare billing.

12. Conclusion

This Healthcare Analytics project demonstrated an end-to-end data analytics workflow using Python, SQL, and Microsoft Power BI to transform raw healthcare data into structured analysis, meaningful insights, and an interactive dashboard.

Python was used to inspect, clean, and prepare the dataset for analysis. This included removing duplicate records, validating billing values, converting date fields, and creating additional variables such as Length of Stay and time-based admission fields. Exploratory Data Analysis was then conducted to understand patient demographics, medical conditions, admission patterns, hospital stays, and healthcare billing.

SQL extended the analysis by answering structured analytical questions using techniques such as aggregation, filtering, conditional logic, Common Table Expressions (CTEs), and window functions. The analysis provided further insight into patient age groups, healthcare billing, admission types, medical-condition rankings, and admission trends over time.

Microsoft Power BI was used to bring the analysis together in an interactive Healthcare Analytics Dashboard. The dashboard combines key performance indicators, analytical visualisations, time-series analysis, and interactive slicers, allowing users to explore patient volumes, healthcare costs, Length of Stay, medical conditions, admission patterns, insurance providers, and test results. This represents the final stage of the project's analytical workflow.

Overall, the analysis identified relatively balanced distributions across several areas of the synthetic dataset, with comparatively small differences in healthcare billing and Length of Stay across different categories. These findings should be interpreted within the limitations of synthetic data and should not be considered representative of real-world healthcare populations or hospital performance. The project demonstrates the practical application of data cleaning, exploratory analysis, SQL querying, data visualisation, and business intelligence within a complete analytics workflow, progressing from raw data to a final interactive reporting solution.

13. References

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